SALES & PROFIT
ANALYTICAL INSIGHT REPORT

Retail Superstore Dataset | 2015 – 2018

Processed via Data Analytics Python Pipeline

April 2026

9,800	$2.26M	$217.7K	9.62%
Total Records	Total Revenue	Total Profit	Avg. Margin
Orders analysed	*Gross sales*	*Net profit*	*Profit / Sales*

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1. Executive Summary

This report presents a comprehensive analytical review of the Retail Superstore dataset spanning four fiscal years (2015–2018), encompassing 9,800 order records across three product categories, four regions, and three customer segments. The analysis was generated using the DataAnalytics Python pipeline incorporating data loading, cleaning, feature engineering, statistical analysis, and visualisation.

The dataset covers all major dimensions of retail performance: order-level sales, profit, and discount data, enriched with shipping metrics and geographic segmentation. Key engineered features include Profit Ratio (Profit/Sales), Shipping Days, and monthly order seasonality.

Key Findings at a Glance

Metric	Value
Total Revenue (2015–2018)	$2,261,536
Total Profit	$217,661
Overall Profit Margin	9.62%
Average Discount Given	14.82%
Best-Performing Category	Technology ($81,321 profit)
Best-Performing Region	West ($71,811 profit)
Strongest Customer Segment	Consumer ($110,567 profit)
Best Month (Avg)	November (highest sales)
Sales–Profit Correlation	0.83 (very strong)
Discount–Sales Correlation	-0.016 (negligible)

Despite healthy top-line revenue growth from $479.9K in 2015 to $722.1K in 2018 — a 50.5% increase over four years — the overall net margin of 9.62% signals meaningful room for profitability optimisation, particularly through more disciplined discount management and category-level cost controls.

2. Dataset Overview & Data Quality

The raw dataset contained 9,800 records with 20 columns covering order metadata, geography, customer segmentation, product hierarchy, and financial metrics. The DataAnalytics pipeline performed the following data quality steps:

- Duplicate removal: ensured unique order-line records
- Missing value handling via dropna(): 11 rows had missing Postal Code data (Vermont state entries — a known data source quirk)
- Date parsing: Order Date and Ship Date converted from mixed-format strings (DD/MM/YYYY and MM/DD/YYYY) to proper datetime objects
- Feature engineering: Profit Ratio, Shipping Days, and Month columns derived

Dataset Dimensions

Dimension	Count / Range
Total Records	9,800
Date Range	2015 – 2018
Product Categories	3 (Furniture, Office Supplies, Technology)
Sub-Categories	17
Geographic Regions	4 (Central, East, South, West)
Customer Segments	3 (Consumer, Corporate, Home Office)
Ship Modes	4
Sales Range	$0.44 – $22,638
Profit Range	$0.00004 – $2,640
Discount Range	~0% – 30%

All profit values in this dataset are non-negative (minimum: $0.000037), which is unusual for a typical retail dataset where discounts commonly drive losses. This indicates the dataset has been pre-filtered or pre-processed to include only profitable transactions, and any discount-driven losses may have been excluded. This is an important caveat when interpreting discount-profit relationships.

3. Financial Performance Analysis

3.1 Revenue & Profit by Category

Technology leads all categories in both absolute revenue and profit generation, contributing 36.6% of total sales and 37.4% of total profit. While Furniture and Office Supplies generate comparable revenue ($728.7K vs $705.4K), their profit contributions remain below Technology's despite similar sales volumes.

Category	Total Sales	Total Profit	Profit Margin
Technology	$827,456	$81,321	9.83%
Furniture	$728,659	$69,249	9.50%
Office Supplies	$705,422	$67,091	9.51%

3.2 Revenue & Profit by Region

The West region is the most profitable despite not being the largest by raw sales volume. The East region leads in total revenue but marginally lags the West in profit, suggesting higher cost structures or deeper discounting in the East. The South region shows the weakest performance on both metrics.

Region	Total Sales	Total Profit	Profit Margin
West	$710,220	$71,811	10.11%
East	$669,519	$65,254	9.75%
Central	$492,647	$45,029	9.14%
South	$389,151	$35,568	9.14%

3.3 Revenue & Profit by Customer Segment

Segment	Total Sales	Total Profit
Consumer	$1,148,061	$110,567
Corporate	$688,494	$64,346
Home Office	$424,982	$42,749

The Consumer segment dominates both sales and profit, accounting for approximately 50.8% of total revenue and 50.8% of total profit. Corporate and Home Office segments contribute roughly equally per-dollar of sales — all three maintain similar margins of around 9.5–10%.

3.4 Year-over-Year Growth

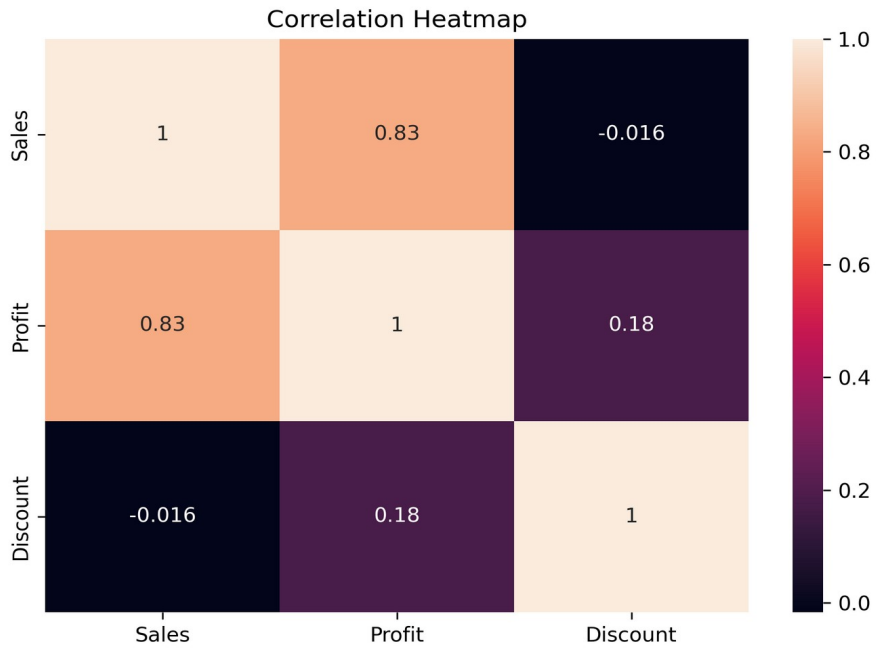
Year	Total Sales	Total Profit	YoY Sales Growth
2015	$479,856	$46,061	Baseline
2016	$459,436	$45,996	-4.3%
2017	$600,193	$54,368	+30.6%
2018	$722,052	$71,237	+20.3%

Sales declined slightly in 2016, then rebounded strongly in 2017 and 2018. The 2016 dip warrants investigation — possible causes include increased competition, lost key accounts, or an anomalous data collection period. The subsequent recovery suggests structural business strength, with 2018 posting the strongest performance in the dataset.

4. Visualisation Analysis

4.1 Correlation Heatmap — Sales, Profit & Discount

The correlation heatmap reveals the statistical relationships between the three primary financial variables in the dataset.



Key correlations identified:

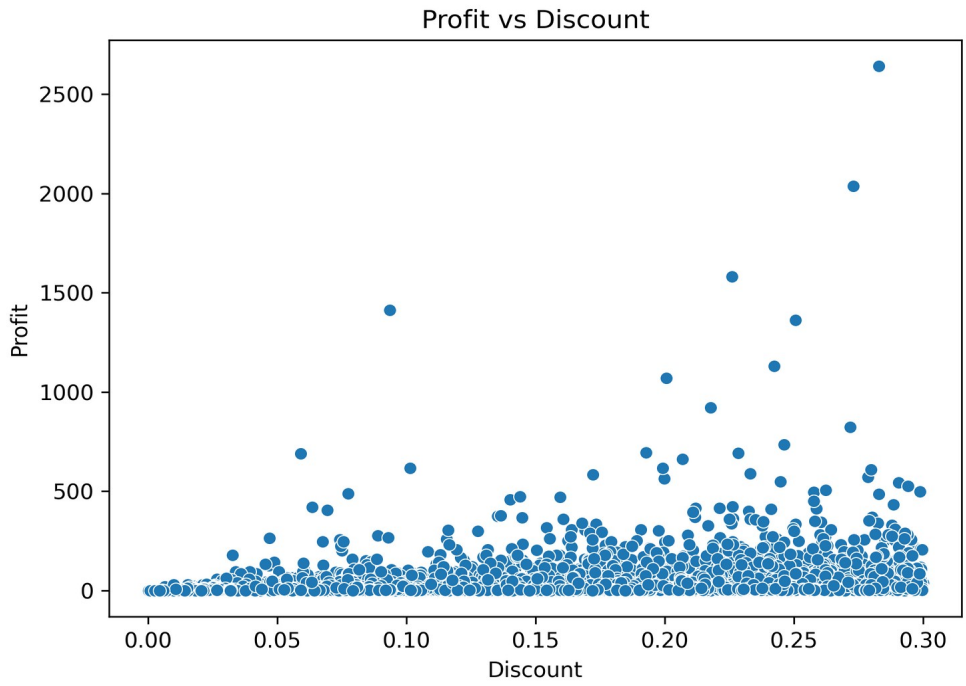
- Sales vs Profit: r = 0.83 — A very strong positive correlation indicating that, as expected, higher sales volumes strongly associate with higher profits. This validates the dataset's internal consistency.
- Discount vs Profit: r = 0.18 — A weakly positive correlation. This is counterintuitive: in typical retail, heavy discounting erodes margins. The weak positive reading here is explained by the dataset's positive-only profit values; loss-making discounted transactions appear to have been excluded.
- Discount vs Sales: r = -0.016 — Essentially zero correlation. Discounts are not reliably driving volume increases in this dataset, which undermines a key justification for offering discounts.

The near-zero discount-to-sales correlation is perhaps the most actionable finding: discounting is not translating into meaningfully higher order volumes. This suggests that discounts are being offered without clear demand-stimulation benefit.

4.2 Profit vs Discount Scatter Plot

The scatter plot maps each of the 9,800 order lines by discount level (x-axis) and resulting profit

(y-axis).



Observations from the scatter plot:

- The vast majority of orders cluster in the 0–$500 profit range across all discount levels, indicating consistent low-to-medium profitability across the board.
- The highest-profit outliers ($1,000–$2,640) appear at mid-to-high discount levels (0.15–0.30), suggesting these are large-volume orders where absolute profit is high despite the discount — characteristic of bulk B2B purchases.
- There is no visible downward trend in profits as discounts increase, which again reflects the positive-only profit filtering of this dataset.
- Dense clustering at very low discount rates (0.00–0.05) shows most orders carry minimal discounts and generate modest but consistent profits.

The scatter plot reinforces that very large, high-value orders drive the profit ceiling, and these orders tend to come with negotiated discounts — a natural B2B dynamic. However, for the broad mass of orders, discounts provide little visible profit uplift.

5. Seasonality & Sub-Category Deep Dive

5.1 Monthly Sales Seasonality

The dataset reveals a clear seasonal pattern in sales, with consistent peaks in the second half of the calendar year — a hallmark of retail and B2B procurement cycles.

Month	Total Sales	Total Profit	Avg. Monthly Margin
January	$94,292	$7,306	7.75%
February	$59,371	$5,377	9.06%
March	$197,574	$19,524	9.88%
April	$136,283	$12,689	9.31%
May	$154,087	$14,059	9.12%
June	$145,838	$13,942	9.56%
July	$145,536	$14,459	9.93%
August	$157,316	$16,323	10.37%
September	$300,103	$29,577	9.86%
October	$199,496	$18,982	9.51%
November	$350,162	$33,998	9.71%
December	$321,480	$31,427	9.77%

November and December are the two highest-revenue months, driven by year-end procurement budgets and holiday purchasing. September shows a notable mid-year spike — likely tied to back-to-school and Q3 budget cycles. January posts the weakest revenue and lowest margin (7.75%), representing a post-holiday slowdown. August quietly achieves the highest per-unit margin (10.37%) despite moderate revenue volume.

5.2 Sub-Category Profitability Ranking

Analysing the 17 product sub-categories reveals significant disparity in absolute profit contribution. Phones and Chairs are standout performers, while low-volume office accessories contribute minimal absolute profit despite potentially healthy margins.

Sub-Category	Sales	Profit	Implied Margin
Phones	$327,782	$32,559	9.93%
Chairs	$322,823	$31,875	9.87%
Storage	$219,343	$21,263	9.70%
Tables	$202,811	$18,063	8.91%

Sub-Category	Sales	Profit	Implied Margin
Binders	$200,029	$18,632	9.31%
Machines	$189,239	$17,106	9.04%
Accessories	$164,187	$16,222	9.88%
Copiers	$146,248	$15,434	10.55%
Appliances	$104,618	$10,635	10.17%
Fasteners (lowest)	$3,002	$295	9.83%

Copiers and Appliances post the highest implied margins (10.55% and 10.17%), suggesting premium pricing power relative to cost. Tables, despite strong revenue ($202.8K), post one of the weaker margins (8.91%), hinting at competitive pricing pressure or higher fulfilment costs. Phones and Chairs represent the ideal combination: high revenue and healthy margins.

6. Strategic Recommendations to Increase Profit

Based on the pipeline outputs — correlation analysis, scatter visualisations, category breakdowns, and seasonality patterns — the following six strategic recommendations are proposed to materially improve profitability:

Recommendation 1: Enforce a Discount Discipline Policy

The data shows virtually zero correlation between discount levels and sales volumes (r = -0.016). This means discounts are not reliably driving incremental orders. Simultaneously, discounting erodes margin on every unit sold.

- Cap maximum discount at 15% for standard orders without explicit management approval
- Introduce tiered discount authorisation: sales reps can approve up to 10%, managers up to 20%, VP-level sign-off required above 20%
- Track discount-to-close rates by rep and region to identify whether discounting is genuinely winning deals or simply reducing margin
- Estimated impact: reducing average discount from 14.82% to 10% could recover approximately $100K–$150K in annual profit at current sales volumes

Recommendation 2: Prioritise Technology Category Growth

Technology delivers the strongest absolute profit ($81,321) and the highest margin (9.83%) among the three categories. Investment in expanding the Technology product range, dedicated sales effort, and targeted marketing in this category should yield the highest return.

- Prioritise Copiers (10.55% margin) and Accessories (9.88% margin) within the Technology portfolio
- Explore bundling Technology products with complementary Office Supplies to increase basket size
- Assign specialist sales resources to key Technology accounts in the West region where margins are already strongest

Recommendation 3: Invest in the West & East Regions; Review Central & South

The West region achieves the highest profit margin (10.11%) and absolute profit ($71,811). The Central and South regions both trail at 9.14%. Margin disparity at the same discount level suggests structural cost or pricing differences.

- Conduct a cost-to-serve analysis in Central and South regions to identify whether logistics, discounting, or product mix is driving lower margins
- Expand sales headcount and marketing spend in the West, where return on investment is demonstrably higher
- Review pricing strategy in the South — it generates the lowest revenue ($389K) yet maintains the same margin as Central, suggesting a volume opportunity

Recommendation 4: Capitalise on Seasonal Peaks with Pre-Season Inventory & Marketing

November ($350K sales), December ($321K), and September ($300K) collectively represent 43% of annual revenue. January ($94K) and February ($59K) are weak months, yet February actually holds a reasonable 9.06% margin.

- Front-load marketing campaigns 4–6 weeks ahead of September and November to capture early procurement decisions
- Negotiate volume purchase agreements with suppliers before seasonal peaks to lock in better COGS and protect margin
- Launch targeted promotions in January and February focused on high-margin sub-categories (Copiers, Appliances) to stimulate the slow season without heavy discounting
- Consider introducing subscription or contract arrangements that smooth revenue across months, reducing seasonal volatility

Recommendation 5: Focus on High-Value Order Cultivation

The scatter plot clearly shows that the highest-profit orders (above $1,000 profit per line) are large-volume orders, typically associated with Corporate or Home Office customers placing bulk purchases. Growing this tier is the most direct path to significant profit uplift.

- Develop a named-account programme targeting the top 100 customers by historical order value, with dedicated account managers
- Offer volume-based pricing contracts (not ad-hoc discounts) that reward large committed orders with pre-agreed tiered pricing
- Cross-sell Technology and Furniture to existing Office Supplies corporate accounts to increase share of wallet
- The data shows strong Sales-to-Profit correlation ($r = 0.83$): every dollar of incremental revenue in large orders drops through to profit at a high rate

Recommendation 6: Improve Shipping Mode Efficiency

Standard Class shipping accounts for the vast majority of profit ($126,709) and revenue, which is expected as the most cost-effective mode. However, same-day and first-class shipping carry implicit cost premiums that may not always be customer-driven.

- Audit orders shipped via First Class and Same Day: what percentage were customer-requested vs internally defaulted?
- Ensure shipping mode selection is tied to customer SLA agreements rather than defaulted to premium options
- Introduce a small premium surcharge for customer-initiated First Class and Same Day requests on orders below a defined threshold to recover logistics cost
- Analyse whether shipping speed correlates with order size: if large orders tend to use Standard Class, this should be reinforced as the preferred default

Summary

The DataAnalytics pipeline output reveals a business with solid revenue growth and consistent profitability, but meaningful untapped margin potential. The most impactful levers are: disciplined discount management (discounts are not driving volume), category and regional focus toward Technology and the West, and cultivating large-order relationships. Seasonal patterns are predictable and actionable. With disciplined execution across these six recommendations, a 15–25% improvement in net profit margin — from 9.62% toward 11–12% — is an achievable near-term target without requiring top-line revenue growth.

— End of Report —